DP4. Sound made from the matter: modal impact sound from shape and material, mass-law walls and reverberation time from distance-field rays, and contact sound from the surface's own roughness field

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Keywords: physically based sound; modal sound synthesis; procedural audio; impact and contact sound; scraping; rolling; Rayleigh-Ritz; Legendre polynomials; modes of a rectangular block; Rayleigh damping; Hertz contact time; oscillator bank; SIMD; occlusion; mass law; transmission loss; signed distance field; reverberation time; Eyring; mean free path; feedback delay network; Minnaert bubble; granular synthesis; surface roughness; texture-driven sound; seeded noise; deterministic audio; temperature-dependent stiffness; shared modes for sound and animation; phase-locked audio; integer phase; destructible worlds; video games; virtual reality.

Classification (CPC): A63F 13/54 (game output involving acoustic signals); G10K 15/02 (synthesis of acoustic waves); G10K 15/08 (arrangements for producing a reverberation or echo sound); H04S 7/305 (electronic adaptation of stereophonic audio signals to reverberation of the listening space).

Related disclosures in this series: DP3 (exact integer time and whole-number phases); DP4b (sound radiation per mode by matched solves); DP5 (the physel: one record for what is seen, touched and heard).

1. Field

Real-time sound for simulated worlds, made from the shape, material and motion of the simulated matter instead of from recordings.

2. Problem and known methods

A world in which anything can be broken, carved, melted or moved needs sounds for shapes that did not exist when it was built. Most games play recordings chosen by material and size. A recording is a real sound, but of another event: it cannot follow the size, the contact point, the temperature or the room, and it repeats.

Known method	Examples	What it lacks here
Modal synthesis: damped modes excited by contact forces	Adrien 1991; van den Doel and Pai 1998; van den Doel, Kry and Pai 2001; O'Brien, Cook and Essl 2001; O'Brien, Shen and Gatchalian 2002	Modes from a finite-element solve of each mesh, ahead of time: a new fragment has none
Fragment soundbanks; modes fitted to recordings	Zheng and James 2010; Lloyd, Raghuvanshi and Govindaraju 2011	Limited to what was prepared
Damping fitted per material	Ren, Yeh and Lin 2013	Constants only
Rayleigh-Ritz modes of a block	Demarest 1971; Ohno 1976; Visscher et al. 1991	A measurement method; not, as far as we found, a run-time table giving any box its tones
Contact spectra; acceleration noise	Johnson 1985; Zheng and James 2011; Chadwick, Zheng and James 2012	Offline, or per precomputed object

Known method	Examples	What it lacks here
Cheaper synthesis of many modes	Bonneel et al. 2008; Raghuvanshi and Lin 2006; Tsingos, Gallo and Drettakis 2004; Phillips 1996	A fixed merge distance removes the slow beats ringing metal keeps
Contact sound from surface profiles and visual textures	van den Doel, Kry and Pai 2001; Rath and Rocchesso 2005; Avanzini, Serafin and Rocchesso 2005; Conan et al. 2014; Picard, Tsingos and Faure 2008; Schmidt and Dionisio 2008; Ren, Yeh and Lin 2010; Chan, Tymms and Colonnese 2021	2D maps keyed to texture coordinates, which broken or carved surfaces lack; noise keyed to time does not repeat when sliding back
Bubbles	Minnaert 1933; van den Doel 2005; Moss et al. 2010	—
Occlusion by rays	Schissler and Manocha 2011; commercial audio engines	Per-surface factors, not the thickness, density and frequency crossed
Reverberation from rays	Krokstad, Strøm and Sørsdal 1968; Savioja and Svensson 2015; Antani and Manocha 2013; Schissler and Manocha 2018; Jot and Chaigne 1991	A mesh ray tracer beside the renderer's own data
Radiation per mode	James, Barbič and Pai 2006	Out of scope: our own method is DP4b

Together they lack: tones for any new piece at once and at no cost per piece; one source of truth for look, motion and sound; wall loss that follows real thickness, density and pitch; the same tones and draws from the same event on every machine; and a rule that keeps each simplification honest.

3. Summary

Every solid is made of small solid primitives (physels, DP5), each with a box size and a material. A struck physel rings with its box's free modes: a Rayleigh-Ritz solve on Legendre polynomials, made once for 12 × 12 proportions and stored as 20 dimensionless frequencies each (11,520 bytes), scaled at the hit by the material's shear-wave speed and the box's size, damped by Rayleigh constants and filtered by the Hertz contact time. Walls take each tone's mass-law loss from the mass per area that rays through the world's distance field cross; rooms set a delay-network reverberator from a reverberation time that rays through the same field estimate. One double-precision vibration solve per kind of object may give its sound to 29.4 kHz and its sway; pitches follow stiffness at temperature; law-driven motion carries sound recipes evaluated from its own integer phase; scraping reads the seeded roughness field that draws the surface. Every random draw is seeded, and a simplification stays only when a reference and a listening test show it inaudible.

4. Detailed description

Numbers marked **derived** are computed from the stated inputs.

4.1 What runs where

Place	Work
Physics (CPU or GPU)	Reports each impact, one per physel per step (the fastest): position, speed, physel id, material, half-sizes, mass, and what it hit (ground, moving object, or resting physel and its material); each moving object's fastest sliding speed and its speed; particle collisions per 40 cm block and sound class
Game thread, once a frame	Picks hits, makes their tones (§4.3-4.4), sends them through a lock-free single-producer queue of 1,024 commands (full: the newest is dropped and counted)
GPU, once a frame	Wall rays (§4.7); every 6th frame, room rays (§4.8); read back two frames later
Audio thread (real time)	Voices (§4.5), particle textures and scrapes (§4.9), the reverberator, a limiter; never allocates, waits or logs. 48 kHz stereo f32, 10 ms periods

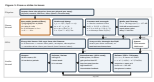


Figure 1. From a strike to tones: the game thread reads the box table, scales, damps and filters the tones and waits two frames for the wall rays; the audio thread rings them and feeds the reverberator, set by the room rays.

4.2 The material's sound values

Per material: Young's modulus E (f32, Pa), density ρ (f32, kg/m³), Rayleigh α and β (2 × f32; 1/s, s), absorption per bounce in three bands (3 × f32: below 800 Hz, 800 Hz-8 kHz, above 8 kHz) and a particle sound class (u8: water, sand, rubble, lava, none). Poisson's ratio is 0.3 for all. A 256-entry GPU table holds (density, absorption ×3) as four f32 per material number: 4 KB.

Material	E (GPa)	ρ (kg/m³)	α (1/s)	β (s)	Absorption	Damping from
Wood	11	650	2.14	3.08×10^{-6}	0.11, 0.07, 0.06	Ren, Yeh and Lin 2013
Steel	200	7,800	6.3	2.12×10^{-8}	0.20, 0.07, 0.06	Same
Glass	70	2,500	18.3	1.43×10^{-7}	0.06, 0.03, 0.02	Same
Porcelain	70	2,400	0.0374	8.41×10^{-8}	0.01, 0.02, 0.02	Same
Stone	50	2,500	20	2.0×10^{-7}	0.02, 0.04, 0.05	Starting values
Brick	15	2,000	40	4.0×10^{-7}	0.03, 0.04, 0.07	Starting values
Ice	9	900	10	1.5×10^{-7}	0.01, 0.015, 0.025	Starting values
Soil, leaves	0.1	300-1,500	300-500	1×10^{-4}	0.20, 0.50, 0.60	Starting values (no ring)

Absorption comes from published room-acoustics tables (Vorländer 2008) and open-source presets.

4.3 Box modes: the table

The solve. For a free, isotropic box with half-sizes h scaled so the largest is 1, the displacement is $u_i = \Sigma\, c_{i,lmn}\, P\hat{}_l(x/h_x)\, P\hat{}_m(y/h_y)\, P\hat{}_n(z/h_z)$, i = 1..3, over $l + m + n \le 10$ (286 terms a component, 858 unknowns), with normalised Legendre polynomials $P\hat{}_l = \sqrt{((2l + 1)/2)}\, P_l$. The 1D integrals of $P\hat{}P\hat{}$, $P\hat{}'P\hat{}$ and $P\hat{}'P\hat{}'$ on [−1, 1] use Gauss-Legendre quadrature with 12 points (exact for these polynomials). Mass: $M = \int \varphi_p\, \varphi_q\, dV\, \delta_{ik}$. Stiffness: $K = \int [\lambda\, \partial_i\varphi_p\, \partial_k\varphi_q + \mu\, \partial_k\varphi_p\, \partial_i\varphi_q + \delta_{ik}\, \mu\, \nabla\varphi_p \cdot \nabla\varphi_q]\, dV$ with μ = 1, $\lambda = 2\nu/(1 - 2\nu)$ (shear-wave speed 1 at unit density). Each 3D integral is a product of 1D integrals, divided by h per derivative along an axis and multiplied by h for the volume. Eigenvalues of $K\,v = \Lambda\, M\,v$ come from Cholesky ($M = LL^T$), $C = L^{-1}KL^{-T}$, Householder tridiagonalisation and implicit QL, in f64. The six rigid modes are dropped; $\omega\hat{}_k = \sqrt{\Lambda_k}$ for the next 20.

The grid. Sort half-sizes a ≥ b ≥ c. The axes are r1 = b/a and r2 = c/b, each over R = 1, 0.8, 0.6, 0.45, 0.33, 0.25, 0.18, 0.12, 0.08, 0.05, 0.035, 0.02 (cube to 1 : 50); cell (i, j) is the box (1, R_i, R_i R_j). These axes put square sections (r2 = 1) and square plates (r1 = 1) on the grid's edge, so their equal mode pairs are never interpolated against shapes whose pairs split. On the axes b/a and c/a instead, a square bar lies inside the grid, and its two equal bending tones are interpolated from neighbours whose pairs split, so they come out apart.

The layout. f32 T[12][12][20] = 11,520 bytes ($\omega\hat{}$ for c_s = 1, largest half-size 1), made once, offline.

At a hit (with §4.4):

```
box_tones(half, mat, v, tau, seed):              # v: speed of the hit (m/s); tau:
contact time (s)
    a, b, c = sort_descending(half)
    i, ti = cell(clamp(b/a, 0.02, 1));  j, tj = cell(clamp(c/b, 0.02, 1))   # R[i] >=
r >= R[i+1], t in ln r
    for k in 0..19: w[k] = exp(bilinear(ln T[i..i+1][j..j+1][k], ti, tj))  # log
frequency, log ratio
    sort w ascending
    cs = sqrt(E / (2 rho 1.3));  rng = seed * 747796405 + 2891336453     # wrapping
u32
    for k in 0..19:
        omega = w[k] cs / a;  d = (alpha + beta omega^2) / 2
```

```
        share = 0.4 + 0.6 xorshift32(rng)          # drawn for every mode, kept or
not
        if d >= omega: continue                    # overdamped
        f = sqrt(omega^2 - d^2) / 2pi
        if f < 20 or f > 18000: continue
        amp = 0.02 v share H(f, tau) g(f, a) / sqrt(k + 1)
        if within 0.5% of the last kept tone: add amp to it; else keep (f, d, amp),
at most 23
    append the thump: f_t = clamp(0.5/tau, 40, 4000), decay 4 f_t, amp 0.08 v min(a /
0.1, 3)
```

Log-log interpolation is exact for power laws (a bending frequency goes as the thickness). Thinner than 1 : 50 uses 1 : 50; rounded physels use their box (a limit of the table; §4.10 solves exact shapes). **Scaling:** $\omega_k = \hat{\omega}_k c_s / A$, A the largest half-size, $c_s = \sqrt{(E / (2\rho(1 + \nu)))}$; exact for similar shapes. Example (derived): a 10 cm steel cube (c_s = 3,140 m/s) has its first mode at 1.42808 × 3,140 / 0.05 / 2π = 14.3 kHz.

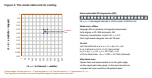


Figure 2. The mode table: 12 proportions on each axis (middle ÷ longest, thinnest ÷ middle), 20 dimensionless frequencies each, interpolated in log frequency and log ratio, then scaled by c_s / A.

4.4 Damping, contact and strength

- **Damping.** Mode k decays at $d_k = (\alpha + \beta\omega_k^2)/2$ per second and sounds at $f_k = \sqrt{(\omega_k^2 - d_k^2)}/2\pi$; $d_k \geq \omega_k$ makes no tone.
- **Contact time** (Johnson 1985): $\tau = 2.87\ (m^2 / (R\ E^2\ v))^{(1/5)}$, *with* $m = m_1m_2/(m_1 + m_2)$, $R^* = R_1R_2/(R_1 + R_2)$, $1/E^* = (1 - \nu^2)/E_1 + (1 - \nu^2)/E_2$, R_1 the struck physel's smallest half-size, v at least 0.01 m/s. The ground is infinitely heavy and large, E = 1 GPa (packed earth). Each tone is multiplied by $H(f) = 1/\sqrt{(1 + (2f\tau)^4)}$: −3 dB at $1/(2\tau)$, −12 dB at $1/\tau$, then −12 dB an octave, so soft or slow hits sound dull. Example (derived): a 1 kg steel ball of 3 cm radius at 1 m/s has τ = 1.4 ms on packed earth (tones fade above about 360 Hz) and 0.19 ms against an equal steel ball (about 2.6 kHz).
- **Thump** (the body's sudden stop): one tone at $f_t = \text{clamp}(1/(2\tau), 40\text{ Hz}, 4\text{ kHz})$, decaying at $4f_t$ per second, with amplitude $4L \cdot v \cdot \min(A/0.1\text{ m}, 3)$.
- **Strength** at 1 m: $a_k = L \cdot v \cdot s_k \cdot H(f_k) \cdot g(f_k)/\sqrt{(k + 1)}$, with L = 0.02 (a calibration constant per m/s); $s_k = 0.4 + 0.6u_k$, u_k from xorshift32 seeded by the physel's record number, one draw per table mode in order (a mode's share never depends on which others are kept); $g(f) = y^2/(1 + y^2)$, $y = f/f_c$, $f_c = 343\text{ m/s} / (2\pi A)$, a simple stand-in for small bodies' weak low tones (radiation is DP4b's subject). Tones outside 20 Hz-18 kHz are dropped; one within 0.5% of the previous adds to it; at most 23 tones plus the thump.

4.5 The strike record and the mixer

Offset	Field	Type	Meaning
0	freq[24]	f32	Frequencies (Hz)
96	decay[24]	f32	Decay rates d (1/s)
192	amp[24]	f32	Starting amplitudes
288	gain[2]	f32	Left, right
296	delay	u32	Samples to wait
300	wet	f32	Share to the reverberator
304	n	u8	Tones used (+3 padding: 308 bytes)

A voice keeps per tone a complex state z and step c (four f32: 16 bytes; 384 for 24 tones), plus gains, share, delay and energy. There are 128 voices.

```
start(strike):  per tone with 0 < 2pi f/fs < pi:  step = exp(-d/fs) e^(i 2pi f/fs);
z = amp + 0i
    energy = (sum amp^2)(gL^2 + gR^2); take a free voice, else the quietest if
```

```
quieter (counted as stolen), else drop
per sample, per live voice after its delay:
    per group of 8 live tones (one SIMD op a line):  z *= step;  acc += Im z
    s = (acc0 + acc4) + (acc1 + acc5) + (acc2 + acc6) + (acc3 + acc7)        # a
fixed order
    L += s gL;  R += s gR;  send += s wet
per block, per voice: skip trailing groups under 1e-10 energy; end it under 1e-9
(about -90 dB) once its delay is over
```

Each tone outputs $a \cdot r^n \cdot \sin(n\omega)$, starting from zero (no click): the classic complex resonator (Smith 2010), whose f32 magnitude drifts about one unit in the last place a step. A limiter follows the volume: e = max(|L|, |R|, e·exp(−1/(0.3 s·f_s))), gain 0.9/e when e > 0.9.

4.6 From the physics to the ears

Each frame, hits from a physel heard in the last 50 ms are dropped, the rest ranked by v·min(A, 0.5 m)/distance, and the 24 loudest played. The listener is the camera (the head in VR): with distance r ≥ 0.3 m, G = volume·√2/max(r, 1 m), x = (direction to the source)·(listener's right), θ = (x + 1)π/4, gains G cos θ and G sin θ; hits wait r/343 m/s.

4.7 Walls by rays through the distance field

The renderer keeps a signed distance field of the resting world for lighting (5 cm voxels, ±20 cm) and one byte a voxel naming the nearest solid's material (255: none). Sound reuses both. Four rays per hit go to the listener: one straight, three moved 8 cm sideways at 0°, 120° and 240° (a pillar does not silence a hit; a hole lets some through). The first and last 30 cm are skipped (the source sits on a surface; the listener may stand by one). Solids count 2.5 cm thicker on each face, so hairline gaps between stacked physels do not leak. The march steps by the field's distance (sphere tracing, Hart 1996).

```
wall_ray(hit q, ray r in 0..3):                      # one GPU thread, 64 a group
    a, b = source[q], listener[q] (moved 8 cm at (r-1)*120 degrees if r > 0);  len =
|b - a|;  d = (b - a)/len
    mass = 0;  x = 0.30
    while x < len - 0.30 (at most 4096 steps):
        s = field(a + d x) - 0.025
        step = min(s > 0 ? max(s, 0.01) : 0.01, len - 0.30 - x)
        if s <= 0: mass += step * density[material_byte(a + d x)]      # 2,000 kg/m3
if none named
        x += step
    result[4q + r] = mass                            # kg/m2
CPU, two frames later, per tone f:  TL_r = clamp(20 log10(mass_r f) - 47, 0, 60) dB
                                    amp *= sqrt(mean over r of 10^(-TL_r/10))
```

The loss is the field-incidence mass law (Beranek and Vér 1992) at each tone's own frequency; the four rays' energies are averaged. Hits wait for their rays: two frames late (22 ms at 90 frames a second). Lasting sounds ask every 10 frames and take the loss at one frequency (a particle class's band centre; 500 Hz for scrapes), silent until the first answer. At most 64 hits (256 rays) go a frame; the rest wait. Example (derived): 20 cm of stone counts as 25 cm, 625 kg/m²: 41 dB at 40 Hz, rising 6 dB an octave to the 60 dB cap from about 360 Hz.

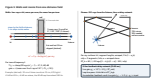


Figure 3. One distance field for walls and rooms. Left: four rays add up the mass per area crossed; each tone loses the mass law's loss at its frequency. Right: 512 listener rays bounce up to four times; their mean free path, escapes and absorption give a reverberation time per band for the feedback delay network.

4.8 Rooms: reverberation time from rays, then a feedback delay network

Every 6th frame, 512 rays leave the listener in Fibonacci directions turned a little each update, bouncing at most 4 times, until they leave the field (open sky), travel 40 m or take 2,048 steps. Each returns 32

bytes.

```
room_ray(i, update):                                  # N = 512
    k = i + 0.5;  z = 1 - 2k/N;  phi = 2.39996323 k + 0.618 update;  d = (sqrt(1-z^2)
cos phi, z, sqrt(1-z^2) sin phi)
    p = listener;  hits = len_hit = seg = total = escaped = 0;  kept[3] = 0
    while hits < 4 and total < 40 (at most 2048 steps):
        if p leaves the field: escaped = 1; stop
        s = field(p)
        if s < 0.03 and seg > 0.1:                    # a surface
            n = normalised central-difference gradient (one voxel);  u = seeded
direction, uniform over n's hemisphere
            d = normalise(0.7 reflect(d, n) + 0.3 u);  hits += 1;  len_hit += seg;
seg = 0
            kept[b] += ln(1 - absorption_b(material at p - 0.03 n));  p += 0.06 n;
continue
        step = max(s - 0.02, 0.02);  p += step d;  seg += step;  total += step
    write (hits, len_hit, escaped, 0), (kept[0..2], 0)
CPU: H = sum hits;  X = sum escaped;  e = X / (H + X);  mfp = sum len_hit / H (10 m
if H = 0)
     per band: keep = exp(sum kept_b / H) (1 - e);  RT_b = clamp(60 / (-10
log10(keep) * 343 / mfp), 0.05, 8) s
               (0.05 s if keep <= 0.001);  level = 0.5 (1 - e);  each update moves
30% toward the new values
```

This is Eyring's formula (Eyring 1930) with the mean free path measured by the rays rather than taken as 4V/S (Kosten 1960), and open sky as a perfect absorber. No mesh is needed: the rays march the walls' field.

The reverberator (after Jot and Chaigne 1991): eight delay lines of 1,031, 1,327, 1,523, 1,801, 2,053, 2,311, 2,633 and 2,903 samples at 48 kHz (prime, 21-60 ms, scaled with the rate). Line i of delay D_i has gain g_i = 10^(−3D_i/RT) and, in its loop, a one-pole low-pass y ← y + (x − y)(1 − a_i), with a_i found by 30 halvings in [0, 0.99] so that its gain at 4 kHz, (1 − a)/√(1 − 2a cos ω + a²), equals 10^(−3D_i/RT_high)/g_i. Feedback is Householder: each line receives (y_i − ¼ Σ y)·g_i plus the input. Left = ½·level·($y_0 - y_2 + y_4 - y_6$); right = ½·level·($y_1 - y_3 + y_5 - y_7$). RT is the mid band's time (0.05-20 s), RT_high the high band's (0.05 s to RT); gains move linearly across a block. Every sound sends 15% of itself to it.

4.9 Particle textures, scraping and rolling

Particles. The particle solver (a material-point method on the GPU) keeps a 32-bit integer sum per 40 cm block and sound class. Each substep a sounding particle older than 0.1 s whose speed jumps by more than gravity could cause (|Δv| − 2gΔt > 0.05 m/s) adds min(jump, 20 m/s) in mm/s, times the fine particles it stands for; integer atomics make the sum order-independent. Once a frame a kernel lists up to 1,024 non-zero (block, class, sum) entries and clears them. The game thread groups them into 1.6 m cells per class; the 16 loudest (sum ÷ distance) get a texture voice of 32 grains at sum (m/s) × grains per m/s ÷ frame time a second (at most 20,000). A grain has pitch f = f_lo(f_hi/f_lo)^u (u uniform), amplitude amp·(0.3 + 0.7u′)·(f_lo/f)^tilt and decay q·f per second; arrivals are Poisson; every draw comes from a seeded xorshift32.

Class	Band (Hz)	q (1/s per Hz)	Tilt	Amplitude	Grains per m/s
Water	330-3,300	0.13	0.5	0.02	1.0
Sand	1,500-8,000	0.6	0	0.008	3.0
Rubble	500-4,000	0.25	0.3	0.025	0.5
Lava	60-400	0.08	0.5	0.05	0.3

Water's band is the Minnaert frequency (Minnaert 1933), about 3.3 kHz × 1 mm ÷ radius, for 1-10 mm bubbles; q = 0.13 gives about eight cycles of ring; the tilt makes big, low bubbles louder.

Scraping and rolling. Sliding (slip > 0.05 m/s) drives at 0.03·√(slip·m·9.81) through a low-pass at 300 + 1,500·min(slip, 4) Hz; rolling (speed > 0.1 m/s, slip < 0.3·speed) at 0.05·speed·√m through 150 + 200·min(speed, 4) Hz; the louder plays; the 8 loudest objects keep a voice holding the body's tones

(box_tones with v = 1, τ = 0.1 ms). Each sample, uniform noise n in [−1, 1) is low-passed, lp ← n + (lp − n)·exp(−2π·cutoff/f_s), and each tone takes z ← (z + lp·level·w_k)·step_k, with w_k = (a_k/|a|)·√(max(d_k, 1)/f_s) scaling out how far noise rings each tone up. Here the noise is keyed to time; §4.13 keys it to the surface instead.

4.10 Exact modes, one solve for sound and sway, temperature, strengths, beats

Targets, a few percent past the best human hearing: frequencies to 29.4 kHz (5% above the highest one published study found audible at very high levels); pitch within 0.083% (1.43 cents); level within 0.24 dB per auditory band (Glasberg and Moore 1990) and 20 ms window; decay within 24% for rings under about 1 s, 9.5% for longer.

1. **One solve per kind of object.** The object's physels (exact rounded shapes, carves) and bonds are meshed into finite elements: one possible choice (an example) is a conforming tetrahedral mesh of the union of the exact shapes minus their carves, refined at curved faces, with bonded faces sharing nodes or joined by springs of the bond's stiffness. $K\varphi = \omega^2 M\varphi$ is solved in f64 for every mode to $2\pi \times 29.4$ kHz (for example by shift-invert Lanczos), refined until no frequency moves more than the pitch target. Supports count: a cantilever rings $(4.730/1.875)^2 = 6.36$ times lower than the free bar. Stored per kind: each mode's ω (f64) and mass-normalised shape, sampled on the surface for sound and at each physel's frame (translation, rotation; f16) for motion: about 6 KB for a tree's sway (50 branch frames × 10 modes × 6 f16) and, as one possible choice (an example), 300-900 bytes a mode for sound (its frequency, decay and 64 surface samples): 77-230 KB for 256 modes. Only unique, struck or dented objects get their own solve.
2. **Sound and sway from the same modes.** The lowest modes drive sway by modal warping (Choi and Ko 2005): each physel gets a translation and rotation, never shear, so it stays exact. The sway is a damped modal law on integer time (DP3), forced by wind (Stam 1997); the high modes make the sound. A tree's sway and its knock share one solve.
3. **A cache, never stored:** keyed to (shape, bonds, material), cleared by a break or dent, solved on worker threads when an object is made or streamed in. Nothing waits on a solve: the box table plays until it lands.
4. **Temperature, wetness, damage.** Frequencies follow $\sqrt{E(T)/E_{ref}}$, E(T) from tables or a model (Varshni 1970). Example (derived): steel at 500 °C keeps about 0.60 of its modulus (EN 1993-1-2), so a bar rings about 0.77 times lower. Wetness and damage change E and damping the same way.
5. **Strengths from mode shapes, on both bodies.** An impulse J along the normal n at x_c starts body A's mode m as $q_m = (J\,\varphi_m(x_c)\cdot n / \omega_{dm})\, e^{(-d_m t)} \sin(\omega_{dm} t)$, and body B's with −n. Taking the Hertz force as a half sine of duration τ (one possible choice (an example)) weights each mode by $|\cos(\omega\tau/2)| / |1 - (\omega\tau/\pi)^2|$ (first zero at $1.5/\tau$), replacing H(f). Small fast pieces add acceleration noise (Chadwick, Zheng and James 2012).
6. **Merging only where the beat dies first:** tones merge only if $|f_1 - f_2| < \min(d_1, d_2)/\ln 1000$, so the pair falls 60 dB within one beat. Derived at 1 kHz from §4.2: steel 0.52 Hz (0.052%), porcelain 0.24 Hz (0.024%), glass 1.7 Hz (0.17%), wood 9.0 Hz (0.90%); a fixed 0.5% (§4.4) is ten times steel's limit. One possible choice (an example): a merged tone takes the summed amplitude and the amplitude-weighted frequency and decay.
7. **Multirate banks:** each tone runs at the lowest rate its pitch needs. One possible choice (an example): banks at f_s/2^j, each tone in the slowest bank whose 0.45 × rate exceeds it, each bank upsampled ×2 by a half-band filter into the next. Banks split over 3-4 workers; the eight-wide step runs in 256-bit SIMD with fused multiply-add where available, 128-bit otherwise, each path repeating itself bit for bit (the paths differ from each other only in rounding). Voices are stolen by loudness at the listener, within the reference's lines (§4.16).

4.11 Law voices: sound on the motion's own integer phase

Motion that follows a closed-form law (a flag's flap, a swinging sign, a fan, a branch's sway) is evaluated from an integer clock: steps of 1/480 s, a fraction, and frequencies of n whole cycles per 2^{21} steps (4,369 s; DP3). A law voice is a sound recipe attached to the law and evaluated on the audio thread from the

same integer phase: the sound stays in step with the picture, with no messages and no drift. Its terms, as one possible choice (an example):

- **Tones:** n (within 1/4,369 Hz of any wanted frequency), an amplitude law, a phase offset: an aeolian tone at the n nearest 0.2 v/d, a fan's blade rate, a mode of the swaying body.
- **Events:** a phase value where a burst plays (a flag's snap as its flap reverses), its sample found exactly from n and the integer time; the burst is seeded by the law's seed and the cycle number, so cycles differ and replays repeat.
- **Noise:** from a counter-based generator, each value a hash of (seed, step, sample within the step): no state, the same bits whenever computed.

```
block anchored at engine time (step0, frac0) at sample s0:          # 100 samples a
step at 48 kHz
per sample s:  (step, f32) = (step0, frac0) + (s - s0) * 480/fs steps     # f32: a
32-bit fraction
    tone (n, A, off):  ph = low 32 bits of ((n * ((step mod 2^21) * 2^32 + f32)) >>
21)    # 128-bit
                       out += A(t) * sine_table(ph + off)
    event (n, at, burst):  if the phase passes `at` before the next sample, start
burst(hash(seed, cycle))
    noise (A, filter):  out += A(t) * filter(hash(seed, step, sample within step))
drawing: ph32 = (n * (step * 2^11 + f11)) mod 2^32
```

The bits agree: at a sample whose time equals a drawn time, the two phases are equal, because $f32 = f11 \times 2^{21}$ and $(step \times 2^{11}) \bmod 2^{32} = (step \bmod 2^{21}) \times 2^{11}$. The audio time is kept to $1/2^{32}$ of a step (4.9×10^{-13} s), so at 10 kHz the fine phase is within about 3×10^{-8} rad of each sample's exact time; sines come from a 4,097-entry table interpolated on 20 more bits, whose worst error is $(2\pi/4{,}096)^2/8 \approx 2.94 \times 10^{-7}$, plus rounding (both derived).

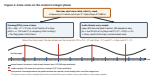


Figure 4. A law voice: one law feeds the drawing (once a frame, 1/2,048-step resolution) and the audio thread (every sample, $1/2^{32}$-step resolution); where their times meet, the phases are the same bits, so a flag snaps as its flap reverses.

4.12 Breaking

A break event carries causes, never sound settings: the bond's position x_b and direction d_b, the force it carried F_b, the released energy and the crack area. The crack's force release excites the parent's modes. One possible choice (an example): each parent mode starts from the deflection the bond held, $q_m(0) = -F_b(\varphi_m(x_b) \cdot d_b)/\omega_m^2$, released over the crack time (crack length ÷ crack speed), which filters high modes as contact time does. Big fragments get exact solves in the background; small ones use a table (after Zheng and James 2010) only where a listening test hears no difference. Each debris cluster gets one granular voice from its collisions; crackle is seeded pops along the crack front; pieces under about 2 cm sound by their rigid-body acceleration.

4.13 One roughness field for the look and the contact sound

A surface's fine relief is a seeded function h(x) of position x in the object's own frame (its look frame), made from the material's roughness spectrum S(k). The field's construction and the sliding model below are one possible choice (an example):

```
make_field(S, seed):            # per kind of object and material; for example J =
64-256 waves, log-spaced in |k|
    per wave j: dir_j = seeded unit vector;  |k_j| = its band's centre;  phi_j =
seeded;  a_j = sqrt(2 S(|k_j|) dk_j)
h(x) = sum_j a_j cos(2pi |k_j| (dir_j . x) + phi_j)
slide, per audio sample:
    x_A, x_B = the contact point in each body's frame, interpolated between physics
steps
    p = h_A(x_A) + h_B(x_B);  p_c = low-pass(p, speed / (2 a_c))     # a patch of
```

```
radius a_c misses shorter waves
    F = max(k_c p_c, -F_N)                                          # never pulls:
the contact opens instead
    mode m of A gets modal force phi_m(x_A).n F;  mode m of B gets -phi_m(x_B).n F
    a_c = (3 F_N R* / 4E*)^(1/3);  k_c = 1.5 (4/3 E* sqrt(R*))^(2/3) F_N^(1/3)
# Hertz
```

The renderer draws wavelengths longer than a pixel's footprint as relief (normals from ∇h) and folds shorter ones into roughness; the sound reads the same h along the contact path at audio rate, each mode obeying $\ddot{q} + 2d\dot{q} + \omega^2 q =$ modal force. As h lives in the object's frame, broken pieces keep their part, crack faces match, and sliding back over a spot repeats its sound bit for bit (on one machine and code path). Relief finer than the stored matter comes from the spectrum, never from noise keyed to time: scraping at 1 m/s to 29.4 kHz needs wavelengths down to 34 μm (derived). Rolling sees the upper envelope of $h(x) - |x - x_c|^2/(2R)$ under a ball of radius R (after Rath and Rocchesso 2005); squeaks use a friction model on the same contact (Avanzini, Serafin and Rocchesso 2005). Pass lines: the field's spectrum within 0.24 dB per band of its material's; scraping within 1 dB per auditory band of the reference's audio-rate contact.



Figure 5. One seeded roughness field, keyed to position in the object's frame, gives the drawn relief and, read along the contact path at audio rate, the force on both bodies' modes: a spot always looks and sounds the same.

4.14 Seeds, never random

Detail too fine to simulate (roughness, bubbles, grains, crackle, scattering) comes from seeded models. Tone shares use xorshift32 seeded by the physel's record number; grains and scrape noise use a seed per voice; room rays use a hash of ray index and update count. Every seed may instead come from the event and the place (one possible choice (an example): a hash of the tick, both bodies' ids and the contact's cell), so an event makes the same tones, shares and noise draws on every run and machine, as shared online worlds need. The mixed audio repeats bit for bit on one machine and code path; on others it differs only in rounding (the fused multiply-add path, §4.10).

4.15 Error bounds and how each is checked

Quantity	Bound	Check
Box modes	Rayleigh-Ritz frequencies are upper bounds, falling as the degree rises	Two degrees compared; a thin bar against Euler-Bernoulli beam theory, which it sits slightly under, as shear makes real bars ring lower; the eigen solver on a matrix with known eigenvalues (a second-difference matrix)
Table interpolation	Exact for power laws in the ratios	A square bar's two bending tones come out equal
Walls	Mass per area to 5 cm voxels and 1 cm steps	Two walls of one thickness and different densities differ by 20 $\log_{10}$ of the densities' ratio
Rooms, reverberator	Against Eyring's formula; the decay asked for against the decay made	§6 works both through
Exact modes, strengths	0.083% in pitch; 0.24 dB per mode	The reference and a listening test (§4.16)

4.16 The rule for simplifications

A simplification (a table instead of a solve, merged tones, stolen voices, fewer rays, a coarser field) stays only when:

1. **A reference** (offline, f64: finite elements for modes, audio-rate contact, wave solves for rooms) finds the error under the thresholds: level per auditory band and 20 ms window, pitch and decay per mode, onsets in 1 ms windows, rooms by ISO 3382-1 (reverberation and early decay times

within 4.8%; clarity and strength within 0.95 dB); and
2. **A listening test** agrees: double-blind ABX (Clark 1982), or MUSHRA (ITU-R BS.1534-3) for statistical sounds without an exact reference, by trained listeners on the best playback available. A listener who reliably tells them apart rejects it (one possible choice (an example) of "reliably": 12 or more right of 16, one-sided binomial $p < 0.05$).

The result is recorded with the simplification. Every sound plays from what is ready at that moment: where an exact result cannot be ready in time, a real-time model plays (very close, slightly imperfect, its error measured by the reference) until the exact one lands. A stale room solve never plays: rays fill in until a fresh one lands.

5. Variants

The method may also:

- **Shapes and solvers:** tabulate other primitives (cylinders, spheres, shells, plates, rods, tapered boxes, rings, superquadrics with roundness as a third axis), layered or anisotropic materials (wood's grain), other Poisson's ratios, supports (clamped, pinned, resting), proportions and mode counts; use other bases (power series, Chebyshev, B-splines, finite or boundary elements), degrees, precisions and interpolations (cubic; tracking modes across crossings); store mode shapes in the table for strengths by contact point; solve boxes at run time, or use a reduced or learned model trained on such solves; build objects' modes from their primitives' by component mode synthesis; apply the exact solve to meshes, voxels, point sets, distance fields or splats converted to solids.
- **Damping and state:** hysteretic or generalised proportional damping, loss tables, fits from recordings with the radiated part removed; frequency and damping that follow temperature, moisture, damage, tension, pressure or the medium.
- **Contact:** nonlinear damped, viscoelastic, plastic or multiple contacts with chatter; Gaussian or measured pulses; soft or layered surfaces; the ground and walls ringing through their own modes.
- **Synthesis:** frequency-domain synthesis, wavetables, or other oscillators (coupled form, state-variable, second-order filters); CPU, GPU, DSP, FPGA or audio chip; any SIMD width and rate (44.1-192 kHz), multirate or not; any voice and tone counts; any loudness or importance measure for stealing.
- **Propagation:** other fields (distance fields at any resolution, sparse or hierarchical; occupancy grids; voxel octrees; ray queries on triangles or primitives), any rays, offsets and thickening; other transmission laws (coincidence, double walls, gaps and leaks, per-band tables, measurements); rooms by Sabine, Millington-Sette, Fitzroy or Arau-Puchades, by image sources or by the energy decay of traced paths; rays scattered by a cosine law, with more bounces, or counting only segments from surface to surface; any number of rays and bands, from the listener or the sources, spread over frames; any reverberator (other networks and matrices, convolution with responses made from the estimate, waveguide meshes, scattering delay networks).
- **Fields and law voices:** fields from Gabor kernels, sparse convolution noise, wavelets, fractal or measured profiles, in 2D or 3D, on any surface parameterisation, shared with touch (haptic) output, friction or wear; law voices on any shared clock (float, fixed point, rational periods) for any periodic or event-driven motion: machines, vehicles, footsteps from gait phases, instruments, swell, rain.
- **Hardware and uses beyond games:** PCs, consoles, phones, headsets, vehicles, servers streaming sound; film and foley; product sound design (how a product clicks, closes or rings, from its design model and material); architectural acoustics; robotics and machine monitoring (expected contact and fault sounds to compare with microphones); tap testing of parts; medicine (training simulators where instruments meet tissue and bone; teaching clinical percussion); musical instruments at their exact pitch; training simulators; teaching.

6. Worked example

Each value is derived from the stated inputs with the formulas of §4.

Step	Inputs	What the method computes
Box modes (§4.3)	A 10 cm steel cube: E = 200 GPa, ρ = 7,800 kg/m³, ν = 0.3	c_s = 3,140 m/s; the first mode, $\omega\hat{}_1 = 1.42808$, at 1.42808 × 3,140 / 0.05 / 2π = 14.3 kHz
Damping (§4.4)	That mode, steel's α = 6.3 /s and β = 2.12 × 10^{-8} s	d = (α + βω²)/2 = 88 /s: the tone falls 60 dB in ln 1000 / d = 78 ms
Contact (§4.4)	A 1 kg steel ball of 3 cm radius at 1 m/s	τ = 1.4 ms on packed earth (tones fade above about 360 Hz); 0.19 ms against an equal steel ball (about 2.6 kHz)
Merging (§4.10)	Two tones near 1 kHz	Merged only within steel 0.52 Hz, porcelain 0.24 Hz, glass 1.7 Hz, wood 9.0 Hz
A wall (§4.7)	20 cm of stone (2,500 kg/m³), counted 2.5 cm thicker on each face	625 kg/m²: 20 $\log_{10}(625 \times 40) - 47 = 41$ dB at 40 Hz, rising 6 dB an octave to the 60 dB cap from about 360 Hz
Two walls (§4.7)	5 cm of wood (650 kg/m³) and of stone (2,500 kg/m³)	Each counts as 10 cm; the loss differs by 20 $\log_{10}(2{,}500/650) = 11.7$ dB at every frequency
A room (§4.8)	A closed 5 × 3 × 5 m room (75 m³, 110 m²), no escapes; mid-band absorption 0.04 (stone) or 0.07 (wood)	With a mean free path of 4V/S = 2.73 m, RT = 60 / (−10 $\log_{10}(1 - \alpha)$ × 343 / 2.73): stone 2.69 s, wood 1.51 s. The ray rules (four bounces from the listener, 30% of each bounce scattered evenly over the hemisphere) give a mean free path of about 2.6 m in this box rather than 4V/S, so the estimate sits a little under the formula with 4V/S
The reverberator (§4.8)	RT = 1.2 s; line 1 of 1,031 samples at 48 kHz (21.5 ms)	$g_1 = 10\hat{}(-3 \times 0.0215 / 1.2) = 0.884$ a pass; every line falls 60 dB in RT, 30 dB in RT/2
Scraping (§4.13)	Sliding at 1 m/s, sound to 29.4 kHz	The field needs wavelengths down to 1 / 29,400 m = 34 μm

7. Figures

Figure	File	Shows
1	`figures/fig1-strike-to-tones.svg`	From a strike to tones (§4.1)
2	`figures/fig2-mode-table.svg`	The mode table and its scaling (§4.3)
3	`figures/fig3-walls-and-rooms.svg`	Walls and rooms from one distance field (§4.7-4.8)
4	`figures/fig4-law-voice-phase.svg`	A law voice on the motion's integer phase (§4.11)
5	`figures/fig5-one-roughness-field.svg`	One roughness field for the look and the sound (§4.13)

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